

CO3 -Assessment tool 3

3. An 8-bit ADC with ± 5 V range reports a quantization noise power of 0.01 V^2 . Verify the calculation and identify the error.

(10)(BL4)(WK4)

MATLAB CODE

```
clc;
clear;
close all;
% Given data
N = 8; % ADC resolution in bits
Vmin = -5; % Minimum input voltage
Vmax = 5; % Maximum input voltage
% Total voltage range
Vrange = Vmax - Vmin;
% Number of quantization levels
L = 2^N;
% Quantization step size
Delta = Vrange / L;
% Quantization noise power
Pq = Delta^2 / 12;
% Reported noise power
P_reported = 0.01;
% Display results
fprintf('ADC Resolution = %d bits\n', N);
fprintf('Input Range = %.1f V to %.1f V\n', Vmin, Vmax);
fprintf('Number of Levels = %d\n', L);
fprintf('Quantization Step Size = %.6f V\n', Delta);
fprintf('Calculated Quantization Noise Power = %.8f V^2\n', Pq);
fprintf('Reported Noise Power = %.4f V^2\n', P_reported);
% Compare values
if abs(Pq - P_reported) < 1e-6
    fprintf('\nThe reported value is CORRECT.\n');
else
    fprintf('\nThe reported value is INCORRECT.\n');
    fprintf('Reported value is %.2f times larger than calculated value.\n', ...
        P_reported/Pq);
end
```

OUTPUT

Command Window

ADC Resolution = 8 bits

Input Range = -5.0 V to 5.0 V

Number of Levels = 256

Quantization Step Size = 0.039062 V

Calculated Quantization Noise Power = 0.00012716 V²

Reported Noise Power = 0.0100 V²

The reported value is INCORRECT.

Reported value is 78.64 times larger than calculated value.